

Funciones modificadas de Bessel

$$\left(\begin{aligned} x^2 \frac{d^2 Y(x)}{dx^2} + x \frac{dY(x)}{dx} - (x^2 + \nu^2) Y(x) &= 0 \\ \nabla^2 \psi - k^2 \psi &= 0 \end{aligned} \right.$$

$$x = -it$$

- $x^2 = (-i)^2 t^2 = -t^2$
- $\frac{d}{dx} = \frac{d}{d(-it)} = \frac{d}{-idt} \Rightarrow x \frac{d}{dx} = (-it) \frac{d}{-idt} = t \frac{d}{dt}$
- $\frac{d^2}{dx^2} = \frac{d^2}{d(-it)^2} = \frac{d^2}{(-i)^2 dt^2} = -\frac{d^2}{dt^2} \Rightarrow x^2 \frac{d^2}{dx^2} = -t^2 \left(-\frac{d^2}{dt^2} \right) = t^2 \frac{d^2}{dt^2}$

Así

$$t^2 \frac{d^2}{dt^2} Y(x) + t \frac{d}{dt} Y(x) - (-t^2 + \nu^2) Y(x) = 0$$

$$t^2 \frac{d^2}{dt^2} Y(-it) + t \frac{d}{dt} Y(-it) + (t^2 - \nu^2) Y(-it) = 0$$

tiene la Forma de la ecuación de Bessel, cuya solución

$$Y_L(x) = Y_L(-it) \equiv I_\nu(x)$$

Puesto que

$$Y_L(x) = I_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s! (s+\nu)!} \left(\frac{-it}{2} \right)^{\nu+2s}$$

$$\Rightarrow I_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s! (s+\nu)!} \left(\frac{-it}{2} \right)^{\nu+2s} = \sum_{s=0}^{\infty} \frac{(-1)^s (-i)^{\nu+2s}}{s! (s+\nu)!} \times \left(\frac{t}{2} \right)^{\nu+2s}$$

$$\text{Pemo } (-i)^{\mu+2s} = \left(\frac{1}{i}\right)^{\mu+2s} = (i)^{-\mu} (i)^{-2s} \\ = i^{-\mu} (-1)^s$$

$$I_{\mu}(x) = \sum_{s=0}^{\infty} \frac{(-1)^s i^{-\mu} (-1)^s}{s! (s+\mu)!} \left(\frac{t}{2}\right)^{\mu+2s} \\ = \sum_{s=0}^{\infty} \frac{i^{-\mu} (-1)^{2s}}{s! (s+\mu)!} \left(\frac{t}{2}\right)^{\mu+2s}$$

$$\text{dado que } x = -it \Rightarrow t = \frac{x}{-i} = ix$$

$$I_{\mu}(x) = \sum_{s=0}^{\infty} \frac{i^{-\mu} (-1)^{2s}}{s! (s+\mu)!} \left(\frac{ix}{2}\right)^{\mu+2s}$$

de modo que

$$I_{\mu}(x) = i^{-\mu} \sum_{s=0}^{\infty} \frac{1}{s! (s+\mu)!} \left(\frac{ix}{2}\right)^{\mu+2s}$$

$$\text{y } J_{\mu}(ix) = \sum_{s=0}^{\infty} \frac{1}{s! (s+\mu)!} \left(\frac{ix}{2}\right)^{\mu+2s}$$

$$\Rightarrow I_{\mu}(x) = i^{-\mu} J_{\mu}(ix)$$

$$i^{\mu} = e^{i\pi/2(-\mu)} = e^{-i\pi\mu/2}$$

$$\Rightarrow I_{\mu}(x) = e^{-\frac{i\pi\mu}{2}} J_{\mu}(xe^{i\pi/2})$$